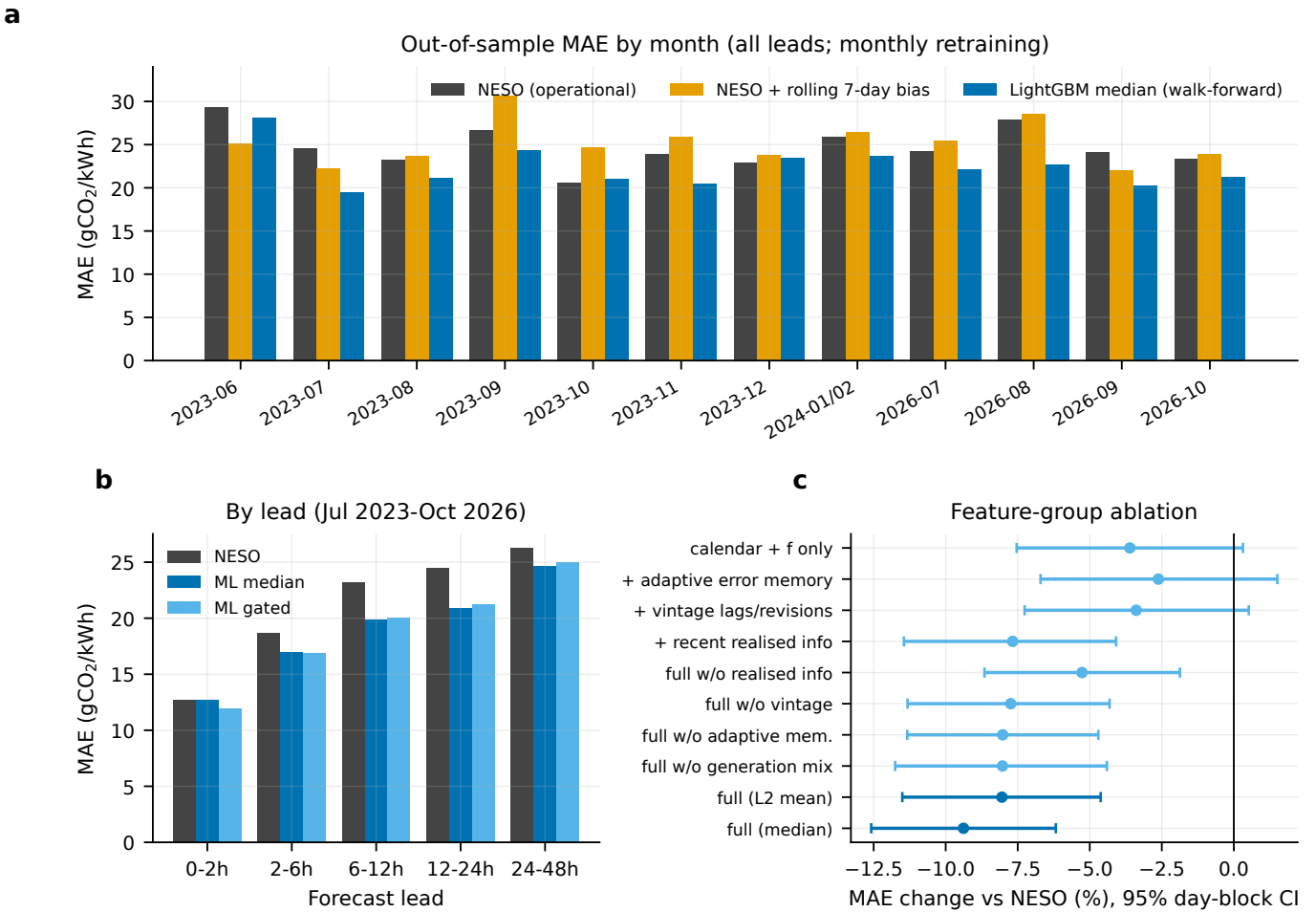
a
Out-of-sample MAE by month (all leads; monthly retraining)
NESO (operational)
NESO + rolling 7-day bias
LightGBM median (walk-forward)
MAE (gCO2/kWh)
0
5
10
15
20
25
30
2023-06
2023-07
2023-08
2023-09
2023-10
2023-11
2023-12
2024-01/02
2026-07
2026-08
2026-09
2026-10
b
By lead (Jul 2023-Oct 2026)
NESO
ML median
ML gated
MAE (gCO2/kWh)
0
5
10
15
20
25
0-2h
2-6h
6-12h
12-24h
24-48h
Forecast lead
c
Feature-group ablation
calendar + f only
+ adaptive error memory
+ vintage lags/revisions
+ recent realised info
full w/o realised info
full w/o vintage
full w/o adaptive mem.
full w/o generation mix
full (L2 mean)
full (median)
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
MAE change vs NESO (%), 95% day-block CI